

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 2 – CONCEPT MAPPING

SET 02

Course Code:	CSA06	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO1 – Precision (BL1, BL2, BL3)	Assessment:	Assessment Tool 2 – Concept Mapping
Weightage:	20% of CIA	Total Marks:	100 (2 Questions x 50 Marks)
CO1 Statement:	<i>Determine algorithm efficiency using asymptotic notations and mathematical techniques including Master Theorem and substitution method. (BL1, BL2, BL3)</i>		
Student Name:	Bestha Akshaya	Reg. No.:	192424336
Section / Batch:		Date:	

Instructions to Students

- Answer BOTH questions. Each question carries 50 Marks. Total: 100 Marks.
- Draw a clear and neat concept map for each question.
- Identify all key concepts and arrange them in a logical hierarchy.
- Show meaningful linkages between concepts using arrows and connecting words.
- Compare related concepts wherever required and justify the relationships clearly.
- Ensure the concept map is well-organized, readable, and properly labelled.

Question Type	Focus Area	Questions	Marks
Concept Mapping	Map algorithm efficiency concepts using asymptotic analysis, search techniques, recurrence relations, and recursion tree method	Q1 – Q2	Q1 –50 Q2 –50
GRAND TOTAL:		100 Marks	

SECTION A – CONCEPT MAPPING

Q1

Consider the concept map: Divide and Conquer → Subproblems → Recurrence → Complexity Map how problem division leads to recurrence. Extend with parameters a and b. Justify their role in complexity. Interpret how this connects design and analysis.

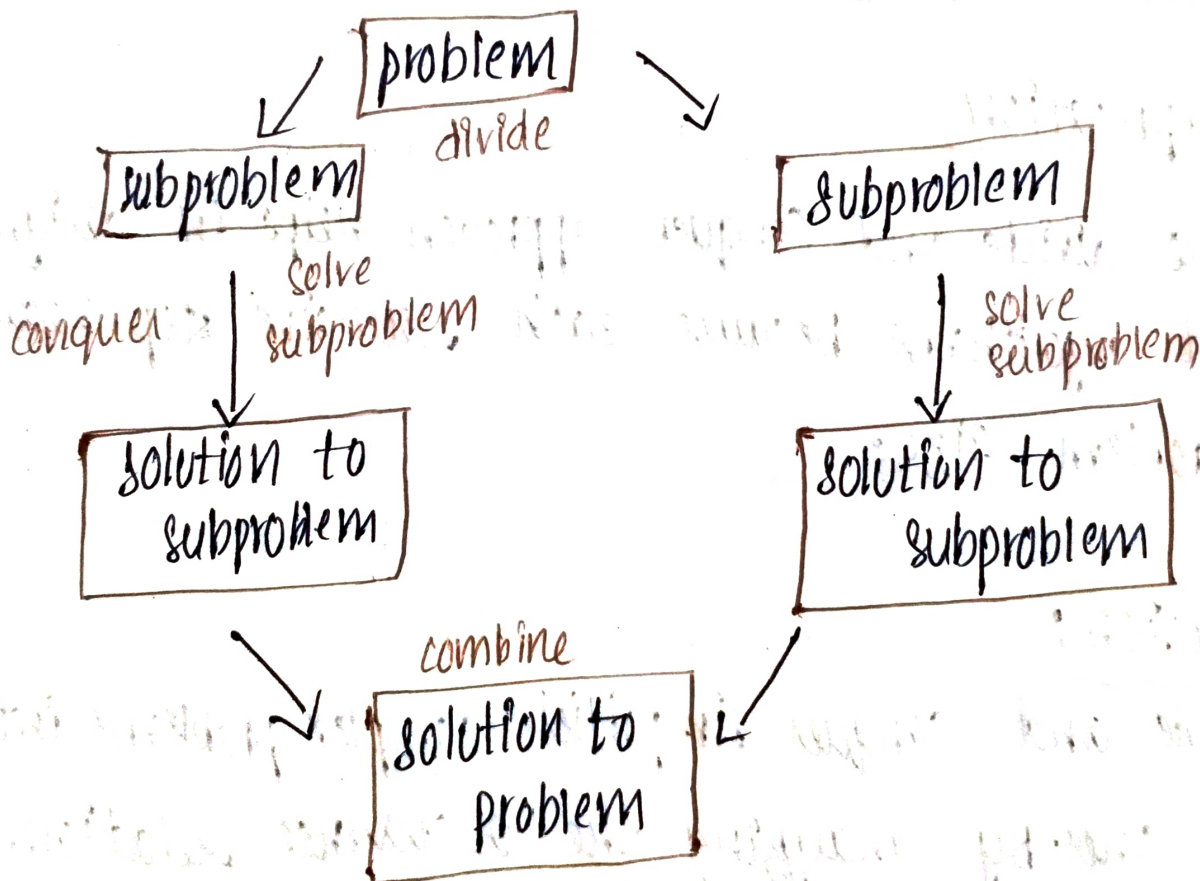
Q2

Consider the concept map: Recursion → Recursion Tree → Level Cost → Total Cost → Time Complexity Explain how recursion expands into a tree structure and distributes cost across levels. Extend the map by including tree height and branching factor. Justify how summing level costs leads to final complexity. Interpret how this visual method aids analysis.

1) Divide and Conquer $\rightarrow$ Subproblem $\rightarrow$ Recurrence $\rightarrow$ complexity

Introduction

Divide and Conquer is an algorithm design technique in which a large problem is divided into smaller subproblem. Each subproblem is solved recursively, and the results are combined to obtain the final solution.



Roles of parameters a and b

The general recurrence relation is

$$T(n) = aT(n/b) + f(n)$$

Where :

- a = No of subproblems created.
- b = factor by which problem size reduced.
- cn = cost of dividing and combining the subproblems.

Explanation :

- The original problem is divided into a smaller problem.
- Each subproblem has size n/b .
- Recursion continues until the base case is reached.

Interpretation

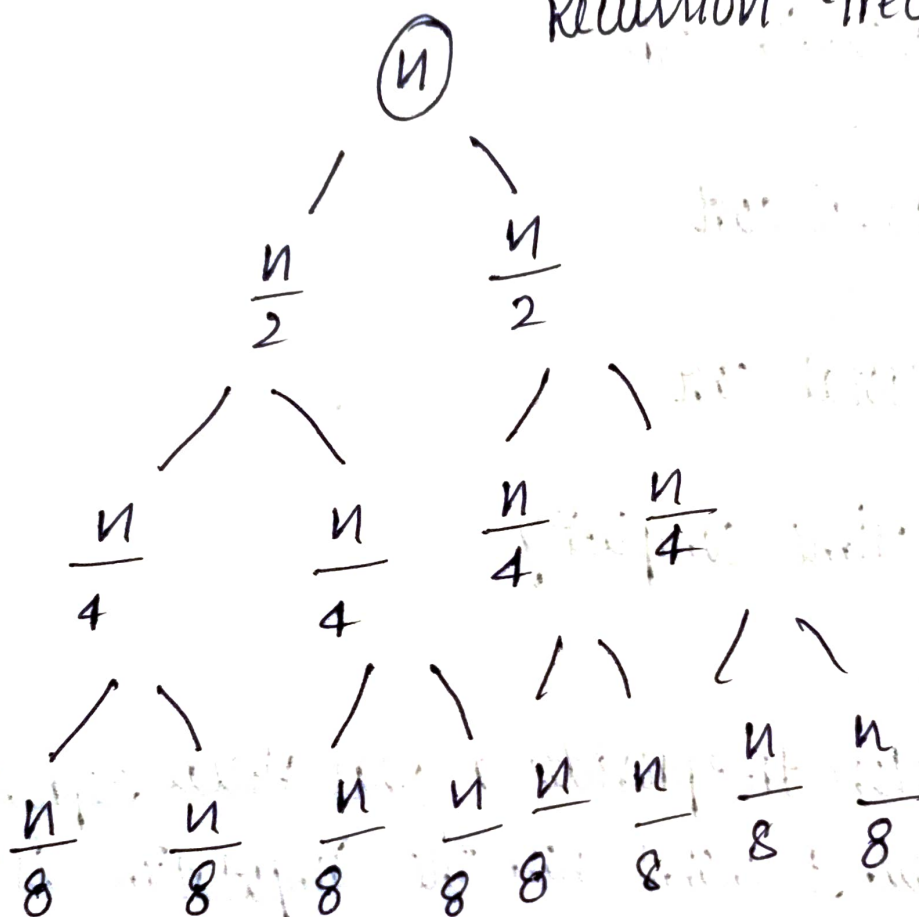
The divide and Conquer approach helps in designing efficient algorithms because each recursive step reduces the problem size.

Conclusion :

Divide and Conquer simplifies complex problems into smaller ones. By analyzing the recurrence relation, the time complexity can be calculated easily, helping us compare the efficiency.

2) Recursion $\rightarrow$ Recursion Tree $\rightarrow$ Level Cost $\rightarrow$ Total Cost $\rightarrow$ Time Complexity

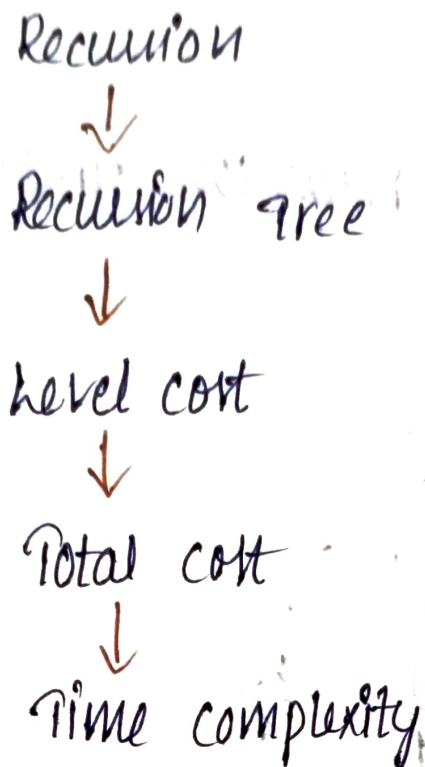
Recursion Trees



Introduction :

The recursion tree method is used to analyze recursive algorithms. It represents every recursive call as a node in a tree and helps calculate total execution cost.

Concept map



Interpretation :

The recursion tree provides a clear visual representation of recursive calls. It shows how the computation is distributed across different levels.

Conclusion :

The recursion tree method is an effective technique for analyzing recursive algorithms. By considering the tree height, branching factor, and total cost.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Identification	25%	Identifies all key concepts from literature accurately and comprehensively	Identifies most key concepts with minor omissions	Identifies limited concepts; some are irrelevant or missing	Fails to identify essential concepts
Linkages	25%	Establishes clear, meaningful, and accurate relationships between concepts	Shows correct relationships with minor gaps	Relationships are weak, unclear, or partially incorrect	No meaningful relationships established
Organization	20%	Concepts are well-structured hierarchically with logical flow and grouping	Mostly organized with minor structural issues	Some organization but lacks clear hierarchy	No logical organization or structure
Integration of Literature	20%	Integrates multiple studies effectively to show connections and comparisons	Shows some integration of literature	Limited integration; mostly isolated concepts	No integration of literature evidence
Clarity	10%	Concept map is clear, neat, visually structured, and easy to interpret	Generally clear with minor readability issues	Clarity is reduced; difficult to interpret in parts	Poorly presented and hard to understand

Marks Evaluation:

Question No.	Concept Identification (25%)	Linkages (25%)	Organization (20%)	Integration of Literature (20%)	Clarity (10%)	Total Marks (Each – 50)
Q1						
Q2						
Overall Total (100)						

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____